



REDEFINING PRISON TYPOLOGIES IN INDIA

Architecture Thesis Project

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Department of Architecture and Planning
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Thesis Guide

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Introduction

The concept of crime, criminals, prisons and jails have been viewed a taboo. There is a general stigma associated with prisons, especially in India. Architecture has the unique ability to invoke thought, feelings and emotions. Our perception of space subconsciously affects our behavior and temperament.

The research done in the dissertation studied the behavioural outcomes of prisoners and it is implemented in designing a prison during the thesis project and thus developing solutions for redefining the prison typologies in India

About the project

Prisons in India today are the epitome of crude architecture, that are specifically designed to inhibit and torture the criminal mind for even small acts of misdemeanor. It does not allow prisoners any chance for reforming and they succumb to the brutal cycle of incarceration and recidivism.

The idea is to use architecture as a medium to design a prison system that incorporates both incarceration and reforming.

Project Objectives

i.To design a prison that imbibes humane architecture and provides moral and cultural education through recreational facilities such as libraries and meditation spaces.

ii.To design the façade and character of the prison that ensures that it isn't an imposing-built structure on the surrounding community, at the same time fosters a sense of punishment within the prisoner.



3rd

state with the highest crime rate

759 : 1

*Gaurd Staff : Inmates
ratio*

Maharashtra

Prisons horror stories

152.7 %

Occupancy Rate

Site Introduction

• Location

19°02'36.4"N 72°56'51.1"E

• About

The site is located in Mankhurd, in Mandale district, Mumbai

Mankhurd is an eastern suburb in Mumbai.

The township lies on the Harbour line.

• Access Roads

Maharashtra Nagar Road and Johnson Marg



Access Roads



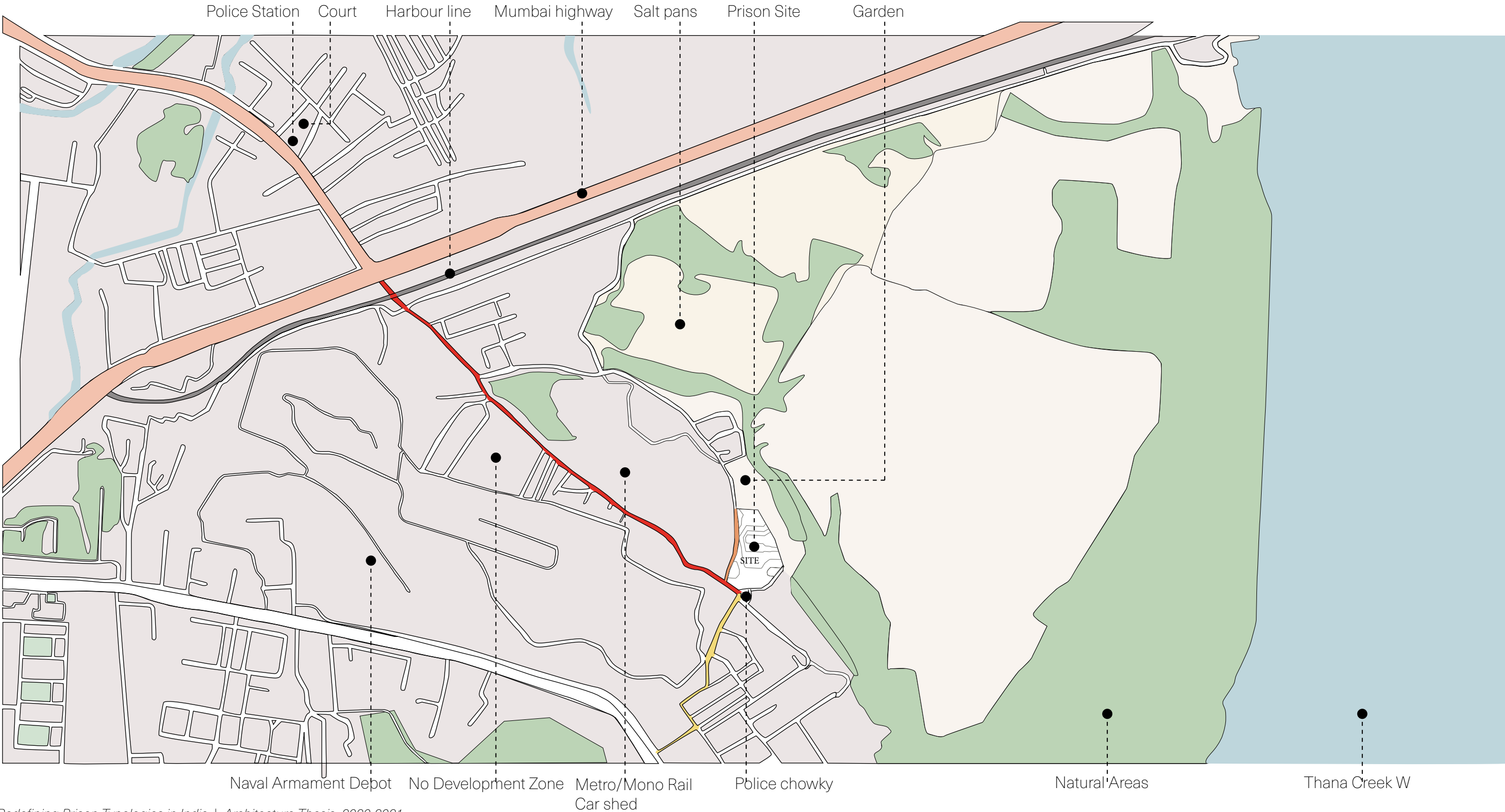
Country



State



Region



Justification

- The site is published in newspapers as a proposed site for jail.
- According to the Draft Development Plan of Greater Mumbai-2034, this area RPU 3.3 is reserved under Public Utilities & Facilities – Correction Facilities.
- The site agrees with the conditions set by Model Prison Manual 2016 for construction of correctional facilities in India.
- The new Mankhurd jail is being built to compensate for the overpopulated Arthur Road Jail. Another problem was that the distance to courthouse was very far (7 km).
- The new Mankhurd jail is 2.7km away from the courthouse.
- In addition there is a police chowky next to the prison site and the nearest police station is approximately 2.6 km away.

SWOT Analysis

Strength

- The areas surrounding the site is occupied by natural areas, a water body on two sides, no development zone, a metro/monorail car shed, a cemetery, thus making sure the site is secluded.
- Irregular shape of the site allows for better security and circulation around the site.

Weakness

- The small size of the site does not allow for a lot of recreational activities

Opportunities

- Additionally, according to DDP there is a police chowki bordering the site. Thus it can provide for easy control and surveillance of the prison.
- There is a new court along with a police station proposed 2.7km away from site

Threats

- NDZ areas could have slum settlement

User Analysis

Based on the space per person designated by the Prison Manual (83.61 sq/m), there should be no more than 440 prisoners.

The social environment of the prison will work on the culture that reinforces opportunity post- incarceration.

Users of the facility will fall under three categories:- Prison inmates, Prison personnel (security and non-security), Visitors (family/friends and law organizations)

Specific Users

- | | | |
|------------------------|----------------------|-----------|
| -Prison Inmates | -Prison Personnel | -Visitors |
| -High Security Inmates | -Gaurds | -Friends |
| -Med Security Inmates | -Kitchen Staff | -Family |
| -Low Security Inmates | -Medical/Councelling | -Lawyers |
| -Undertrial Convicts | -Teaching Staff | -Academic |



Land Development Plan

Surrounding Areas

Railway Station

Mankhurd Railway Station – 0.72 km

Bus Depot

Mohit Patil Nagar – 0.12 km
Railway Salkalp Colony – 0.20 km
Shivaji Nagar Depot – 1.31 km

Police Station

Mankhurd Police Station – 0.4 km
Shivaji Nagar Police Station- 1.60

Hospital

Apex Hospital – 0.13km
Sagar Hospital – 0.22km
Vishwakarna Hospital – 0.4km

Fire Station

Mankhurd Fire Station- 0.4 km

The areas surrounding the site consist of a no development zone, natural areas, metro rail / car shed, naval armament depot, rehabilitation and resettlement areas and a police chowky.

Vegetation



Manila Tamarind



Peepal



Sausage Tree



Vehicular and Pedestrian Traffic Map

The main traffic vehicular and pedestrian is restricted to the Mumbai highway and the roads leading up to the site are isolated, making it suitable for a prison



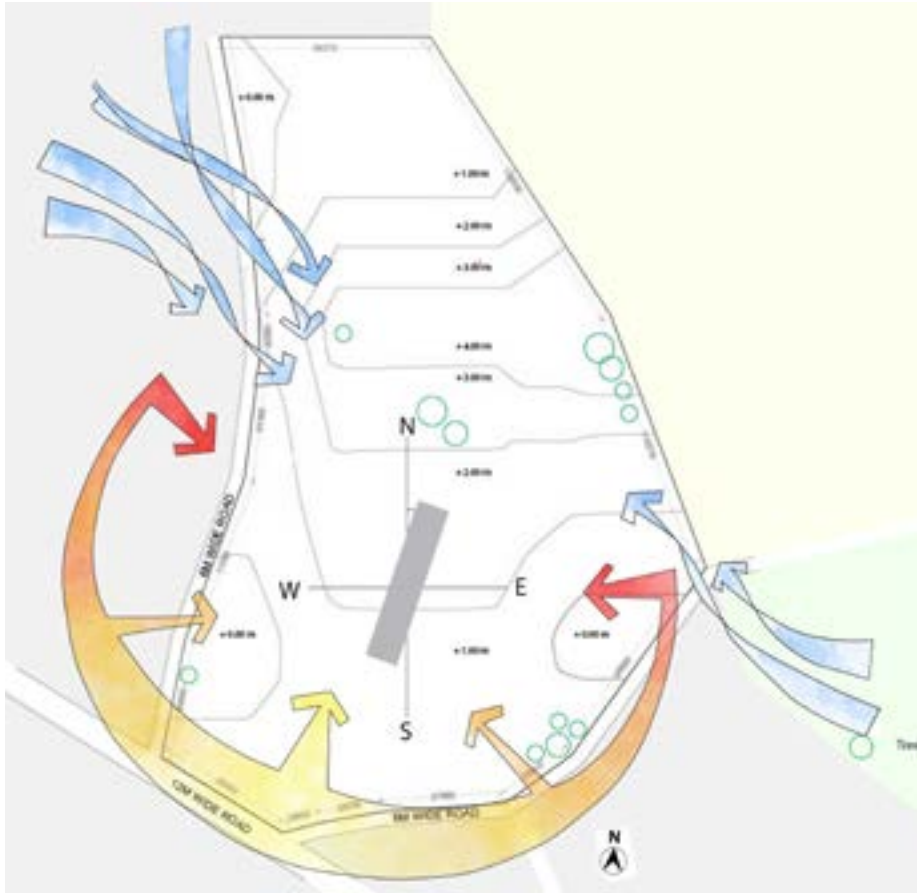
Built- Unbuilt Areas



Green Areas



Surrounding Water Bodies



Site Plan

Climate Analysis

Temperature

The temperature in the **Summer** from March to May is **30-38 degrees Celcius**.
The temperature in **Monsoon** from June to September is **23-28 degrees Celcius**.
The temperature in **Winter** from December to January is **15-25 degrees Celcius**.

Wind and Rainfall

The direction of the wind is predominantly in the **Southwest in the monsoon** and **North west during the rest of the year**.
The maximum and the minimum windspeeds is between 8.8 km/hr and 09 km/hr.
Mumbai experiences round 1500- 2500 mm of rainfall. The rainy season is confined to June to October for around 60 - 70 days .
The relative humidity varies between 41- 67 %

Building Orientation

The South and southwester part of the building have to shaded. The northern facade could consist of glass. The prison rooms should be oriented towards the northeastern side to achieve best climatic orientation.

Site Details

Site Area

50688 m²

Site Area After Boundary Walls

34520m²

Development agency

Maharashtra State Police Housing Corporation (MSPHC)

Total Built up

55500 M2

Total Ground Coverage

50%

FSI

$$1 + 0.5 = 1.5$$

Max Height

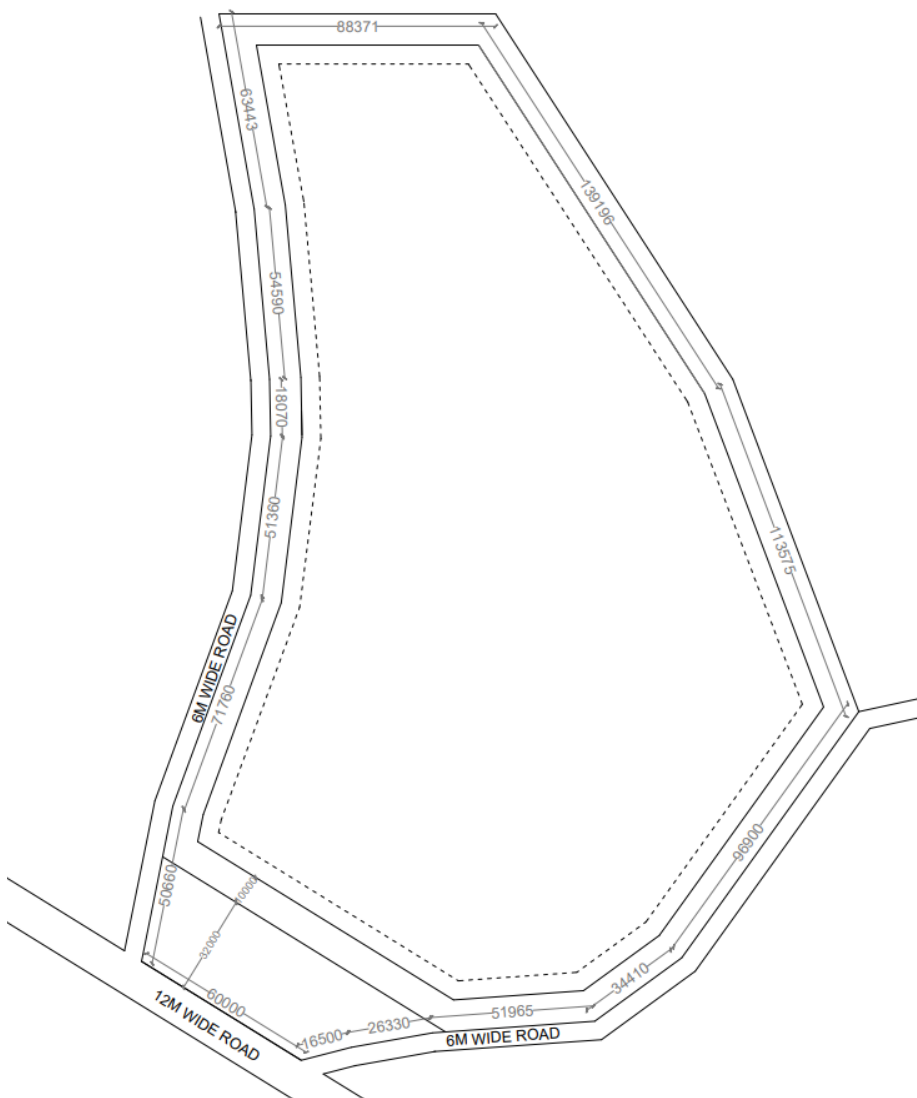
15 m

Parking

One parking space for every 37.5 sq. m of office space up to 1500 sq. m
Parking for Car : 2.5 m²
Parking for Police Bus : 5.5 m²

Set Backs

Front - 42 m setback to incorporate boundary walls and open space + 6m
Sides - 10m (Boundary Wall) + 6m
Back - 10m (Boundary Wall) + 6m



Parameters

Road Network: To access the site, there is a main 12m wide road and two 6m roads that could either be used as roads for services or access to staff housing.

Electricity: Tata power Mankhurd Substation at 1.24 km from site. Tata power Co. has installed a transformer of 500 kVA near the site and a 250 kVA DG set in case of power failure.

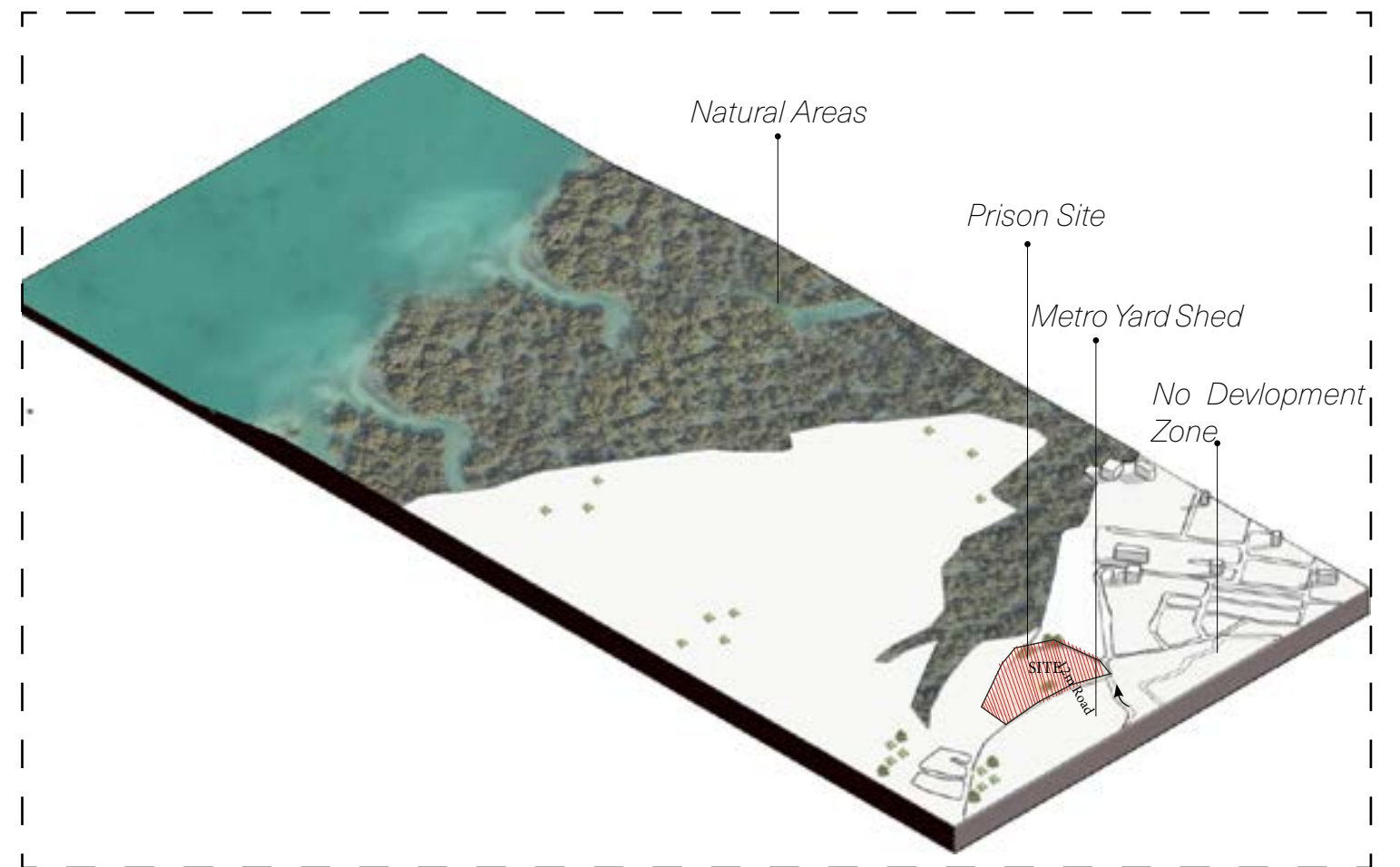
Water Availability: Water for domestic purposes is sourced from the Municipal Corporation of Greater Mumbai. For commercial purposes water from STP which is 3.7 km away from site.

Depth of water table: Pre-monsoon- 2.67m bgl to 4.25 m bgl ,
Post monsoon- 2m bgl to 5 m bgl

Soil type and bearing capacity: Weathered to moderately hard basaltic rock and the soil is marine clay/ slit. This means that all structures on site would have to have pile foundations.

Contours

The site is flat with a few periodic changes in elevation of 4m



Site Axonometric Diagram

REQUIREMENTS

Area Program

S.No	Space	Units	Unit Area	Total Area (m2)
1	Entrance Gate / Control Block			153.5
	Reception	1	42	42
	Visitor Detail	1	7.5	7.5
	Item Storage	1	7.5	7.5
	Search Area	1	7.5	7.5
	Guard Checking	1	7.5	7.5
	Lounge	1	7.5	7.5
	WC	2	4	8
	Vehicle Check	1	6	6
	Vehicle Entry	1	60	60
2	Admin			263.5
	Admin and Prison Processing	1	40	40
	Reception	1	4.5	4.5
	Details	1	9	9
	Pre Processing Holding	1	9	9
	Search Area	1	9	9
	Interrogation Room	1	9	9
	Personal Belonging Storage	1	9	9
	Processing and Bedding	1	15	15
	Post Processing Holding	1	15	15
	Processing Officer	1	9	9
	Officer	1	9	9
	Lounge	1	9	9
	Armory	1	9	9
	Meeting Room	1	9	9
	Lounge	1	15	15
	IT	1	9	9
	HR	1	9	9
	Office	3	9	21
	Briefing Room	1	9	9
	Maintenance	1	9	9
	Security Office	1	9	9
	IT Room	1	9	9
	CCTV / Control Room	1	9	9
3	Visitor Center			276.5
	Exhibition	1	69	69
	Visitor Reception	1	15	15
	Visitor Details	1	44	44
	Guard Room	3	4.5	13.5
	Security Search	1	11.5	11.5

Watch Tower
320m²

	Prisoner Meeting	12	3	36
	Lawyer Meeting Room	5	7.5	37.5
	Conjugal Meeting Room	4	12.5	50
4	Inmate Prison Cells			
a	Low Security Prison Cells			914
	Entrance	1	13	13
	Security Room	1	20	20
	Guard Room	1	13.5	13.5
	Inmate Triple Room	44	13.5	594
	Store Room	1	13.5	13.5
	Common Showers	4	35	140
	Common Toilets	4	13.5	54
	Laundry	1	33	33
	Activity Room	1	33	33
b	Medium Security Inmate Cells			2010
	Entrance	1	14	14
	Security	1	13.5	13.5
	Barber	2	20	40
	Counsellor Room	1	13.5	13.5
	Doctor	2	20	40
	Treatment Room	2	20	40
	Inmate Triple Room	76	13.5	1026
	Common Toilets	8	13.5	108
	Common Showers	8	28	224
	Guard Room	6	13.5	81
	Linen Storage	2	20	40
	Activity Room	6	27	162
	Reading Areas	6	28	168
	Laundry	2	20	40
c	Maximum Security Prison Cells			85
	Guard Posts	3	5	15
	Inmate Room	18	3.5	63
	Inmate Shower	2	3.5	7
d	Undertrial Prison Cells			653
	Entrance	1	31	31
	Guard Room	3	5	15
	Office	3	7	21
	Guard Break Room	3	10	30
	Guard Washroom	3	6.5	19.5
	Storage	3	10	30
	Inmate Room (6 inmates)	9	34	306
	Common Shower	3	34	102

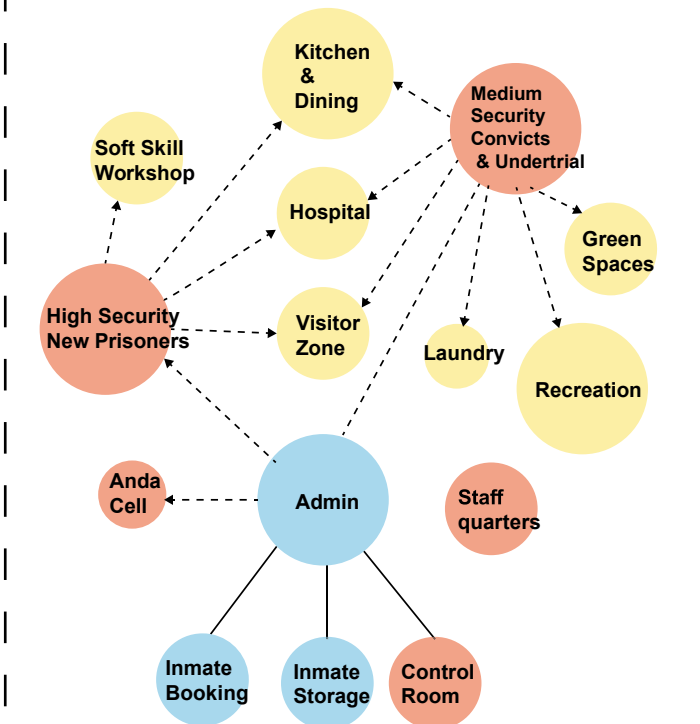
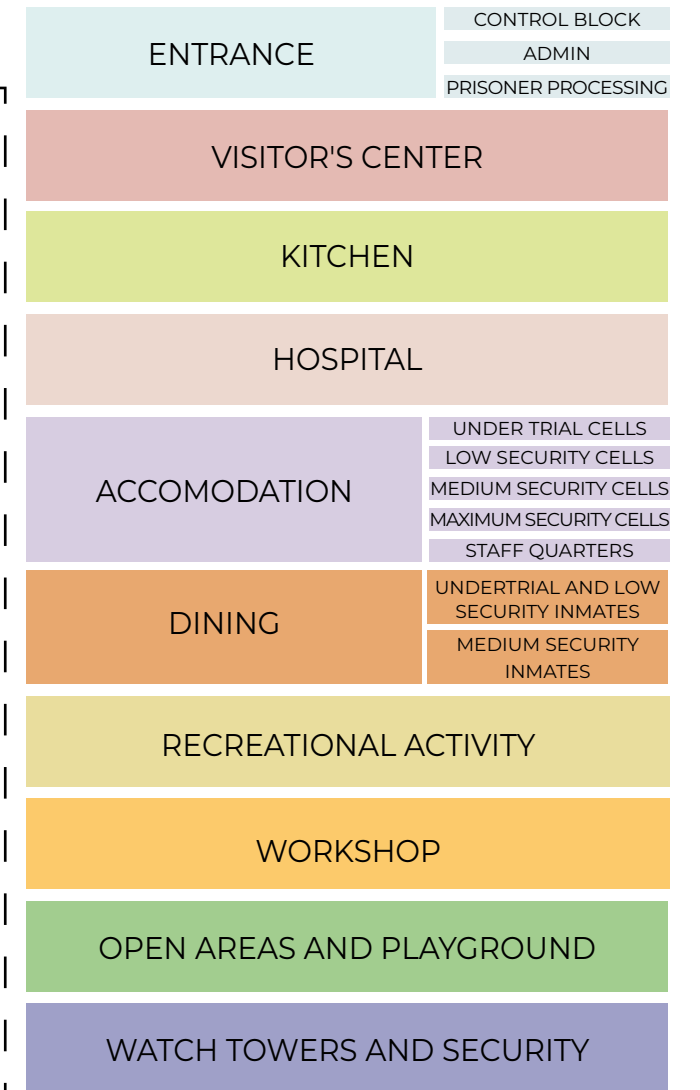
Roads
5328 m²

Built Up Area
15358 m²

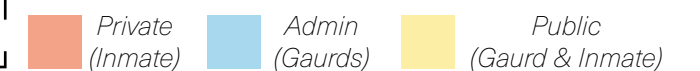
	Computer Room	1	14.7	14.7
	Activity Room	2	42	84
5	Kitchen and Dining			775
	Kitchen Space	1	56	56
	Bakery	1	16	16
	Pantry	1	20	20
	Cold Storage	1	12	12
	Dry Storage	1	12	12
	Fuel Storage	1	6	6
	Storage	1	6	6
	Hand Wash	1	15	15
	Undertrial Dining	1	210	210
	Convicts Dining	1	210	210
	Canteen	1	210	210
6	Clinic			541
	Guard Room	1	2.5	2.5
	Waiting Room	1	25	25
	Doctor's Room	4	10	40
	Toilet	1	6	6
	Dentist	1	25	25
	Pharmacy	1	13	13
	Storage	2	8.5	16.5
	Janitor's Room	2	5	10
	Doctor's Changing	1	4.5	4.5
	AHU	1	4.5	4.5
	Anesthetic Room	1	9	9
	Pre-Op	1	20	20
	Surgery	1	28	28
	Entry (Surgery)	1	4	4
	Wash (Surgery)	1	3.5	3.5
	Sluice (Surgery)	1	3.5	3.5
	ICU	1	14	14
	Post-Op	1	14	14
	General Ward	2	84	168
	Ward- Toilet	3	17.5	52.5
	Isolation Ward	1	100	100
7	Staff Housing			630
8	Recreational /Reformation Activities			2095
9	Sports Block			1821
10	Vocational Block			774

Ground Coverage Area
7348 m²

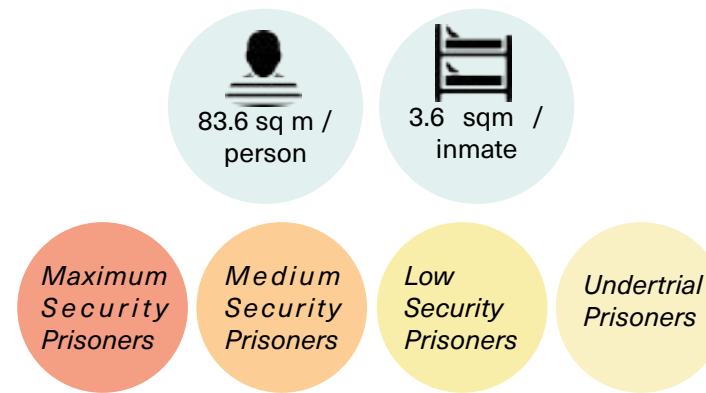
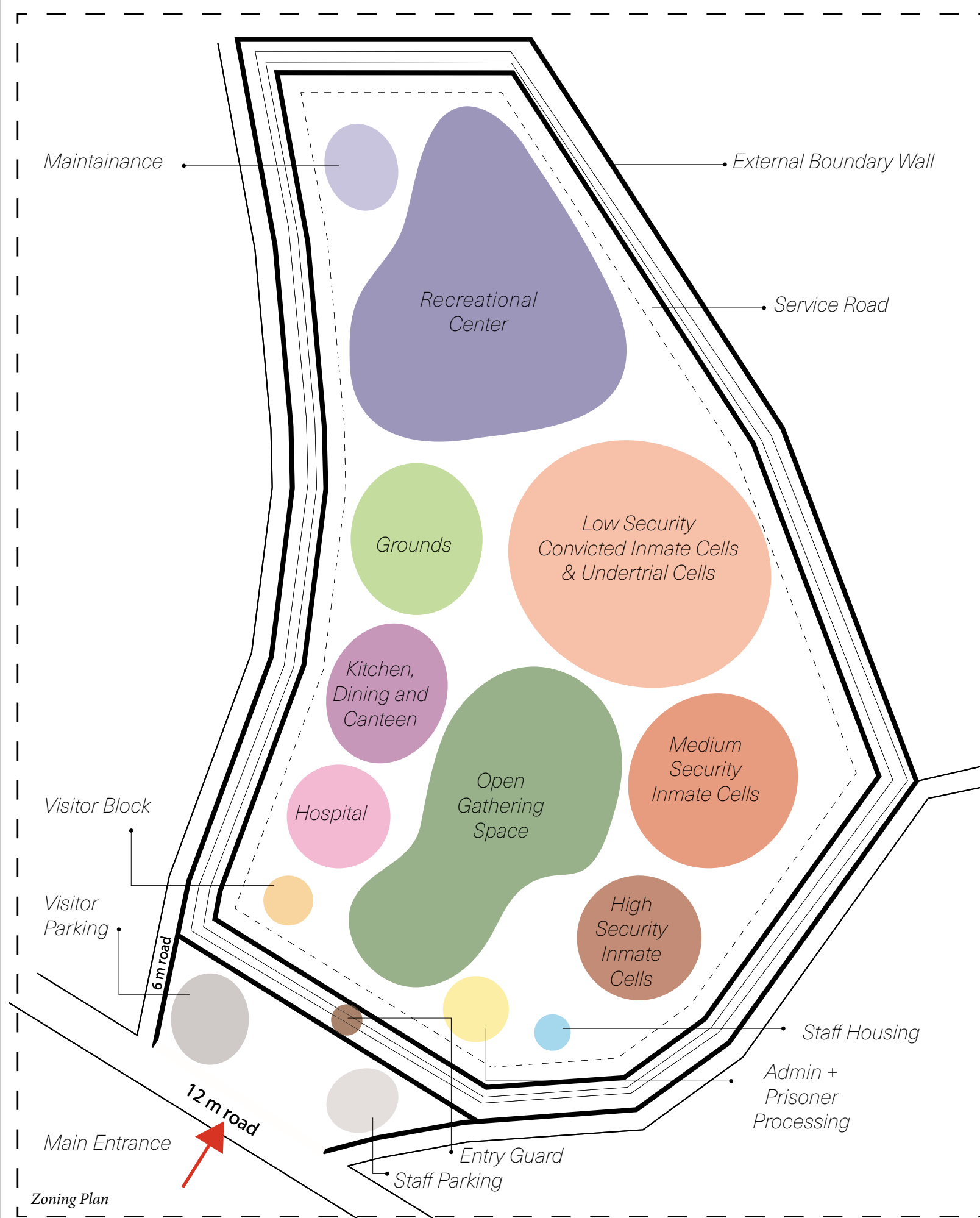
Broad Area Breakup



Bubble diagram showing connection of spaces



ZONING AND CONCEPT



Zoning Concept

The zoning of the prison campus is based two categories

-Prisoner Reformation: Axis of Reformation (Upwards)

-Prisoner Security: Axis of Security (Downwards)

Prisoner Reformation

1 Entry

Maximum & Medium Security Prison Cells

Priority: Healing and Repentance

Access to: Counselling, Clinic

2 Development

Low Security Prison Cells & Undertrial Cells

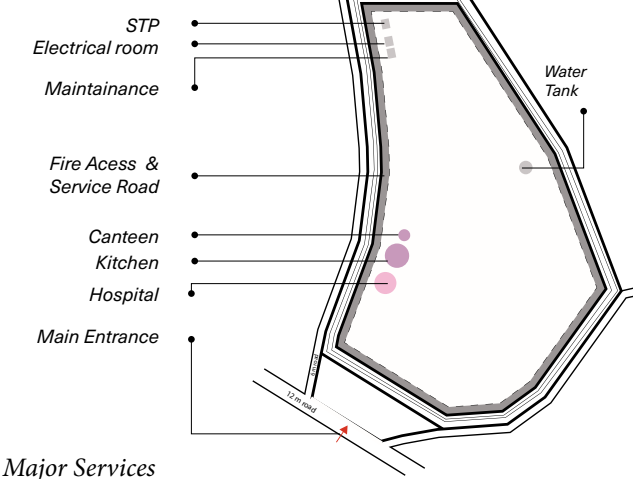
Priority: Healing, Development, Reformation

Access to: Counselling, Workshops, Recreation

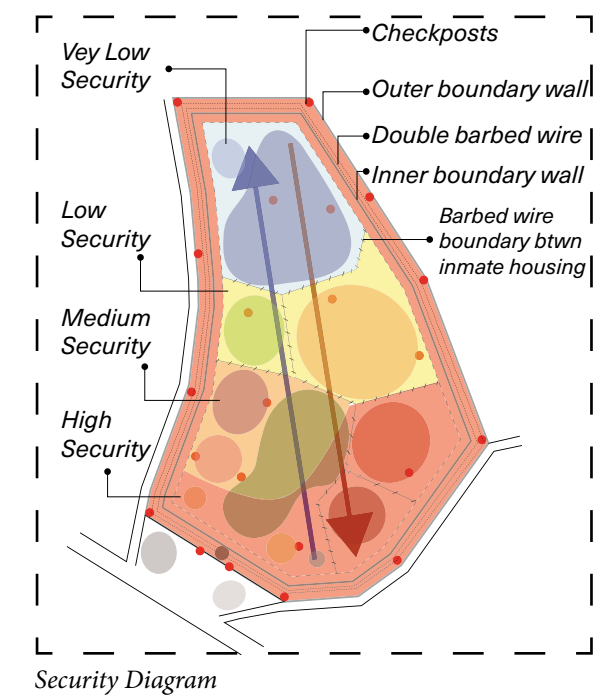
3 Reformation/ Re-integration

Recreational & Sports Complex Facilities, Open Grounds

Services



Security Features



1 Entrance and Exit

Upon entry, there are 4 levels of security to enter the prison campus.

- Outer Boundary wall
- Barbed Wire
- Electric Barbed wire
- Inner Boundary Wall

2 Watch Towers

3 Internal Security

Primary Internal Perimeter

This is a brick wall with jail at intervals that serves two functions: ventilation and primary security

Secondary Internal Perimeter

This contains barbed wire, grill and a cable-based detection system that uses electromagnetic cables attached to the fence to detect noise generated along the fence and prevents escape outside each prison compound type

4 Additional Security

Triangular Monitoring

Vertical Security

Inmates are housed on the basis of their crimes





Site Plan

Legend

- | | |
|-------------------------------------|------------------------------------|
| 1. entrance | 14. medium security convict prison |
| 2. security room | 15. low security convict prison |
| 3. visitor parking | 16. under-trial quarters |
| 4. gaurd parking | 17. football ground |
| 5. control block | 18. sports block |
| 6. visitor block | 19. recreational block |
| 7. admin/ prison processing | 20. gathering grounds |
| 8. hospital | 21. electrical room |
| 9. kitchen&dining | 22. maintenance room |
| 10. vocational block | 23. stp |
| 11. gaurd housing | |
| 12. gallows | |
| 13. maximum security convict prison | |

Activity Pattern

→ Vehicular → Inmate → Visitor → Gaurd



Gaurd Movement



Vehicular, Visitor & Inmate Movement

Recreational

Workshops
Sports Complex
Football Ground
Vocational
Gathering Space

Inmate Cells

Undertrial
Low Security
Medium Security
Maximum Security

Services

Canteen
Kitchen & Dining

Clinic
Visitor Center

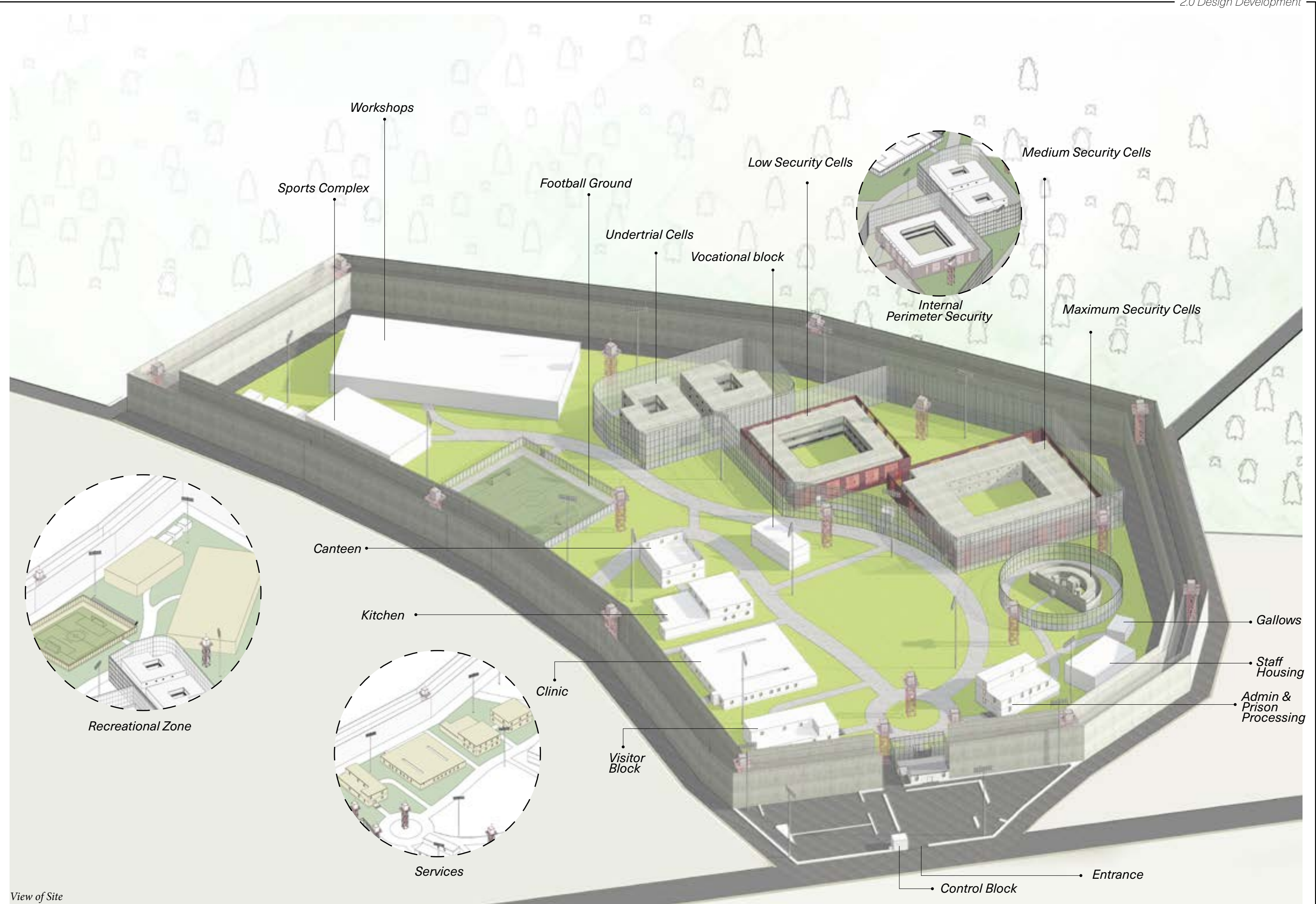
Green Spaces

Major and Minor Road Network

Security

Main Perimeter
Internal Perimeter

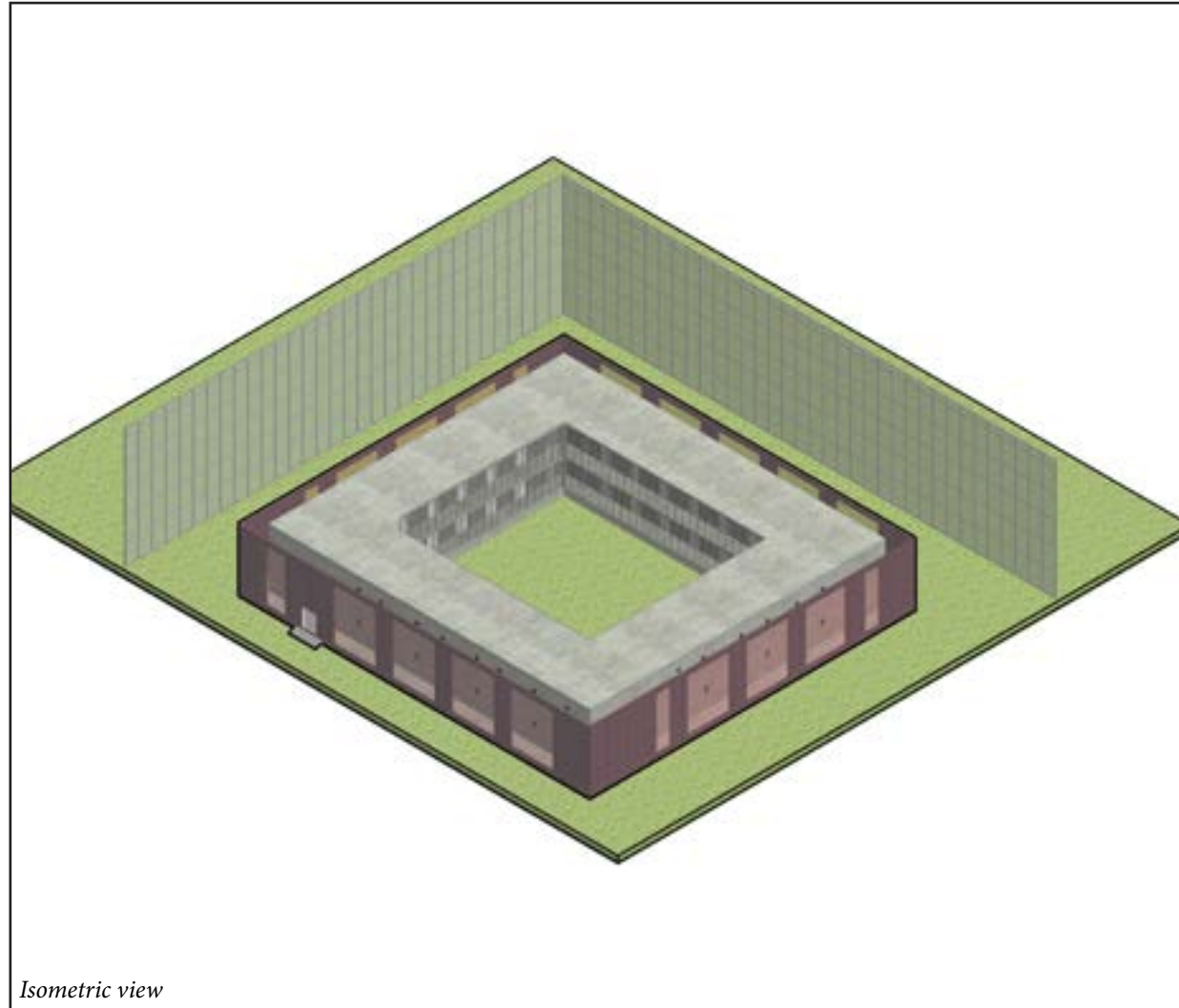
Site



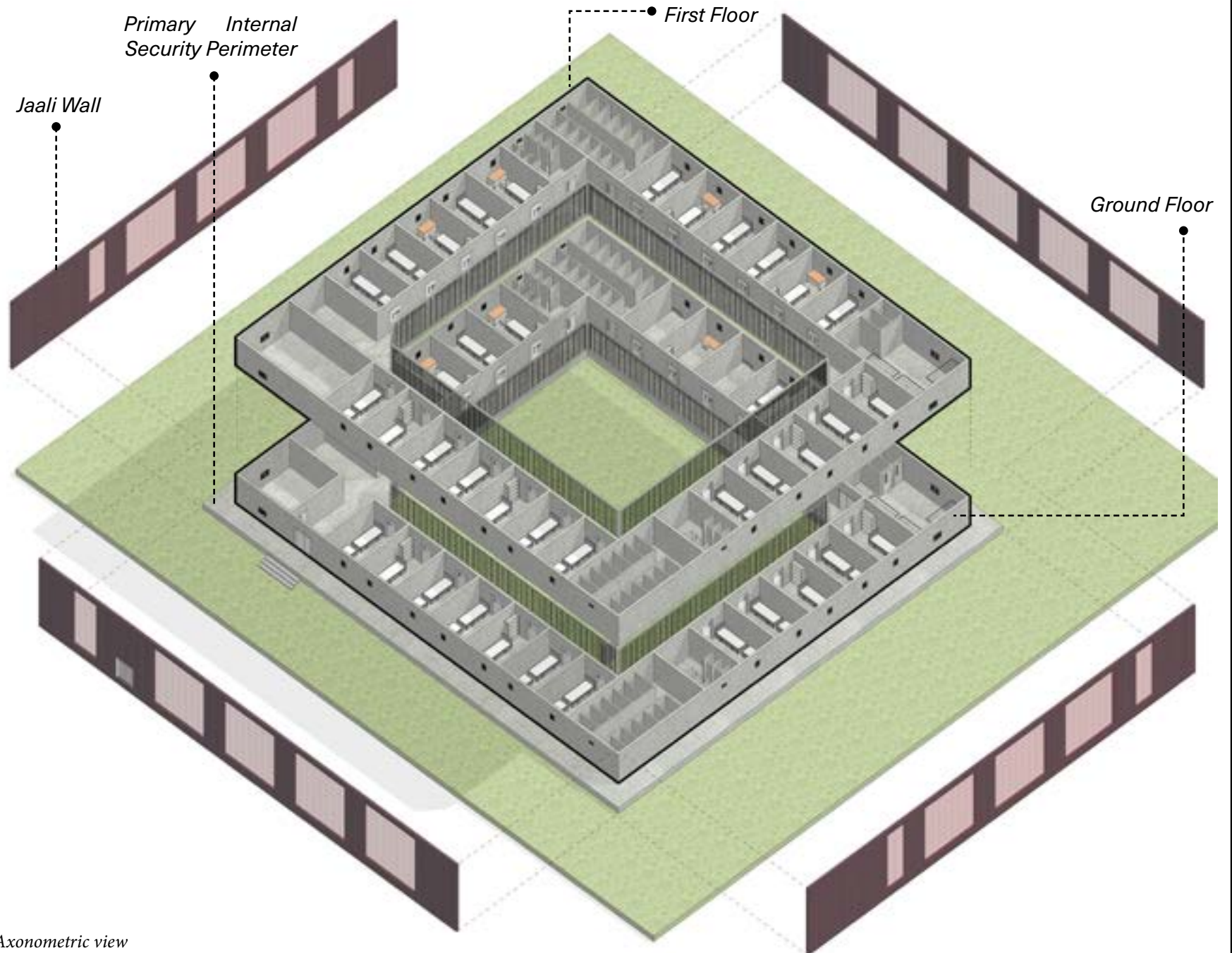
View of Site

LOW SECURITY PRISON CELLS

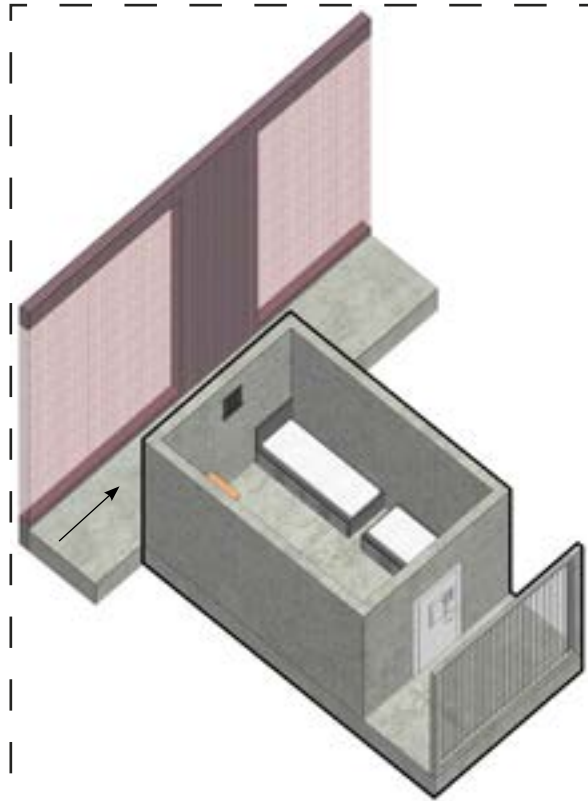
Inmate Housing



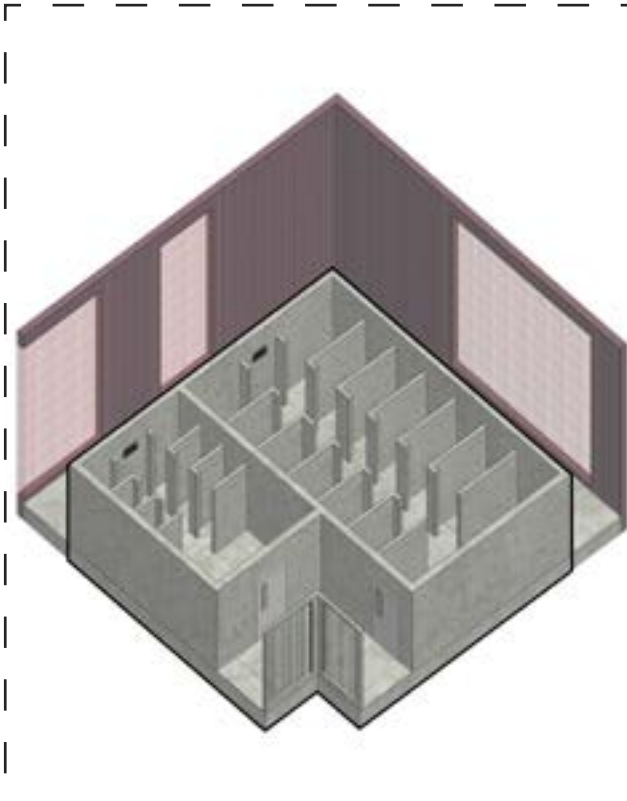
Isometric view



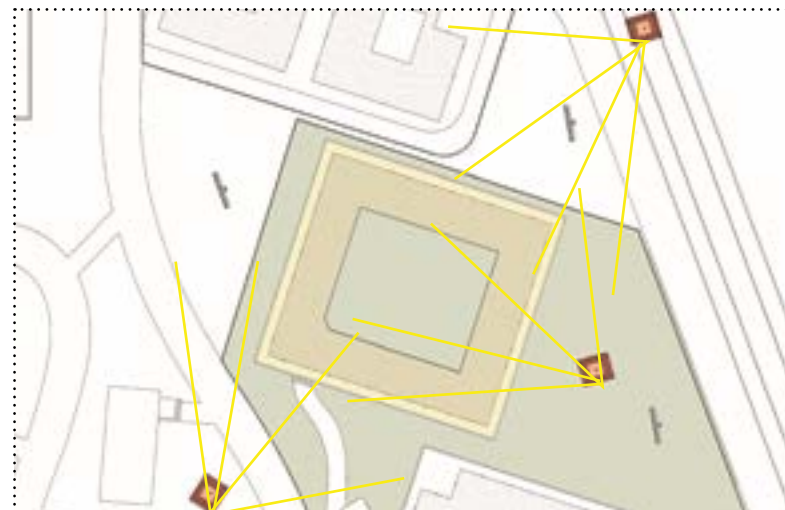
Axonometric view



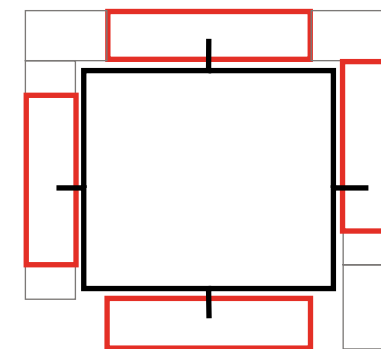
Isometric view of a Room



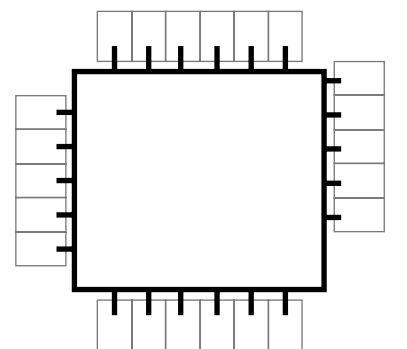
Isometric view of a Common Washrooms



Placement of Watch Towers



Placement of Residential Units



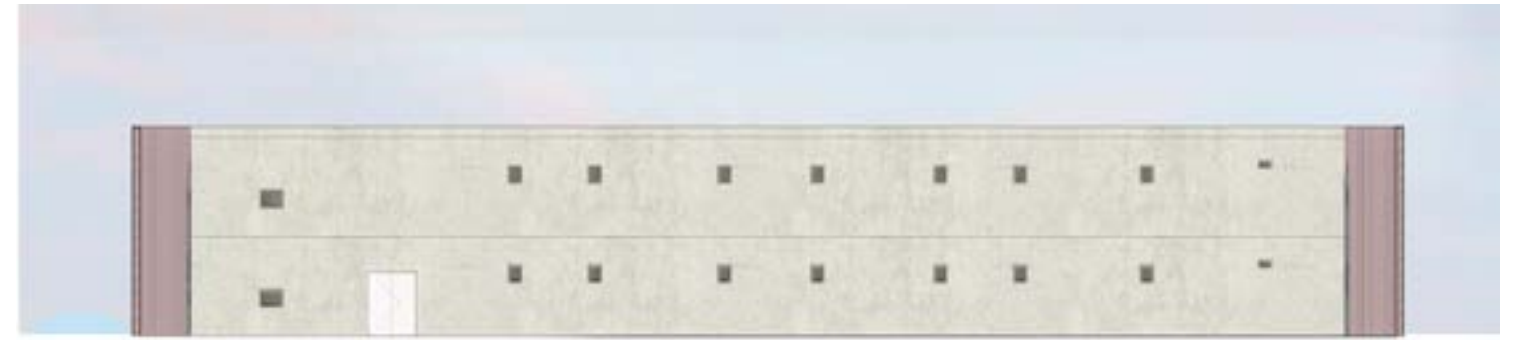
Residential Unit Layout

LOW SECURITY PRISON CELLS

Inmate Housing



Elevation



Sectional Elevation



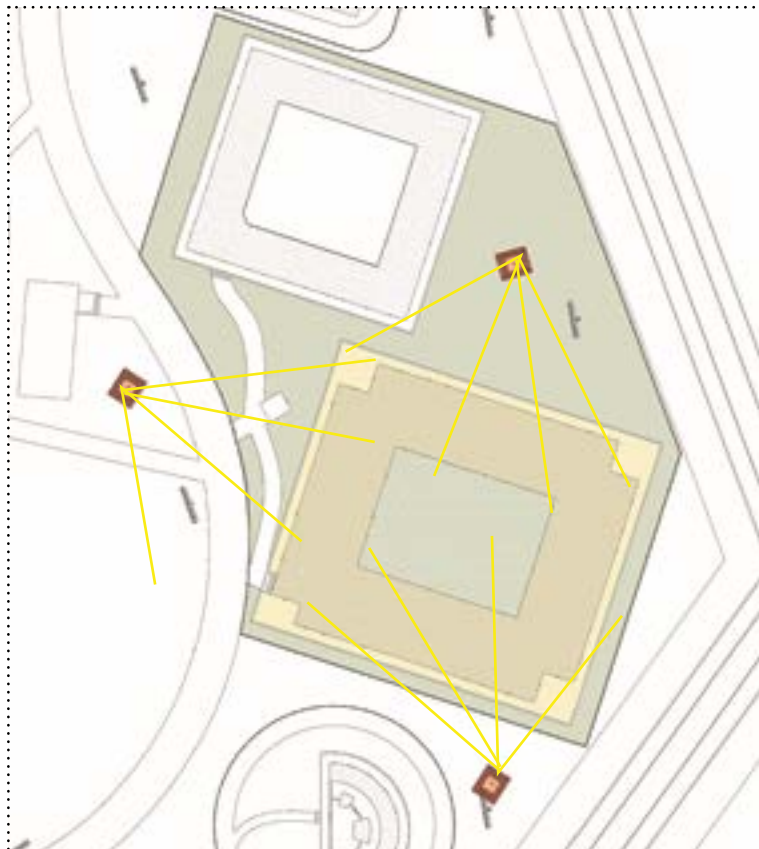
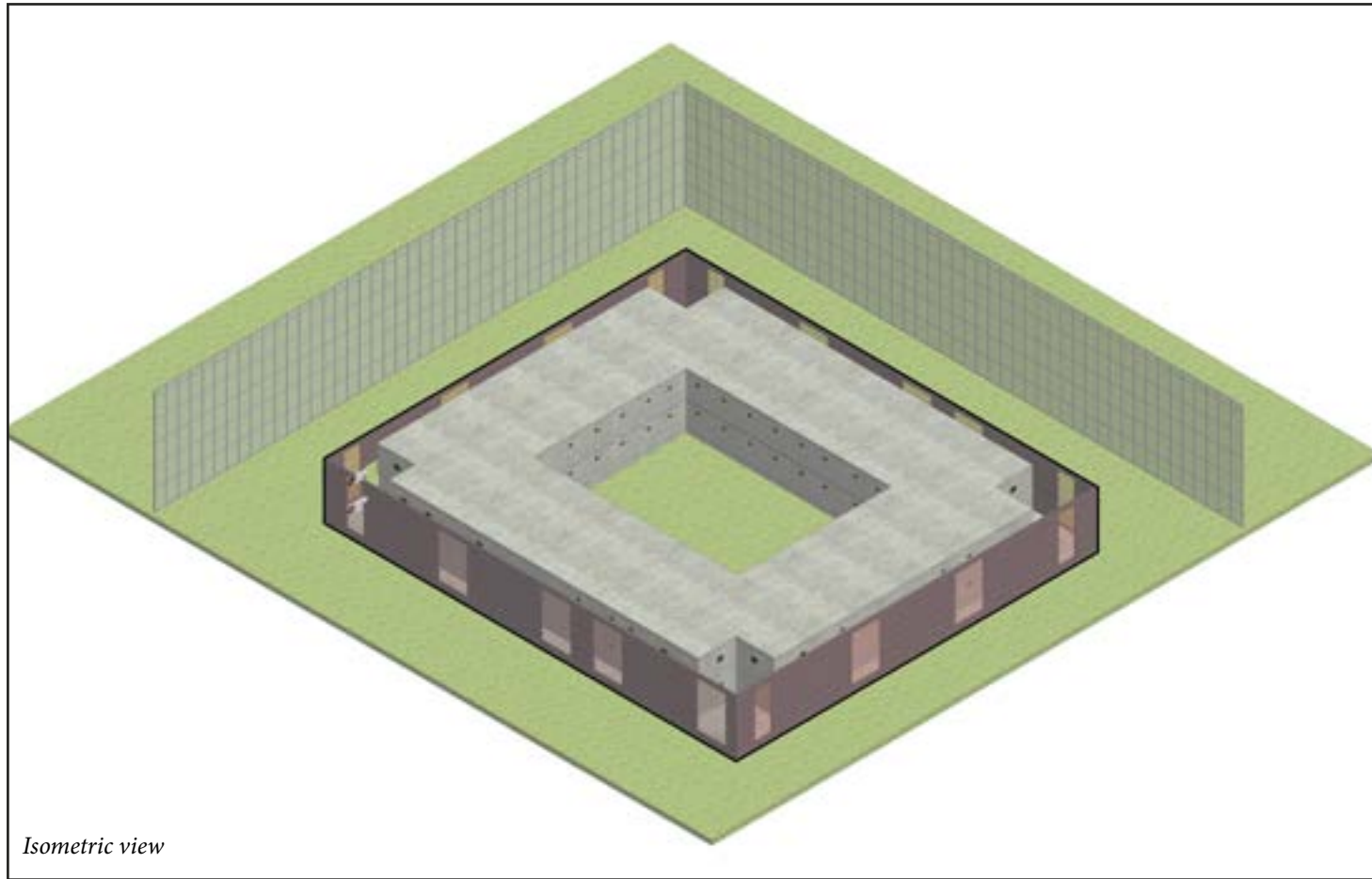
View of Corridor and Courtyard



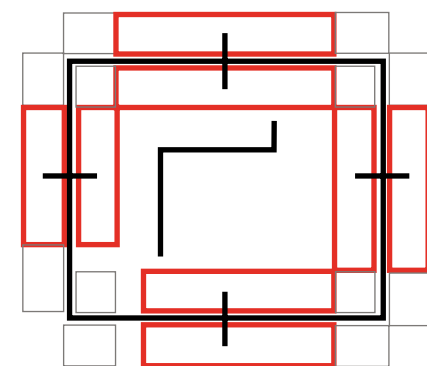
View inside a low security prison cell

MEDIUM SECURITY PRISON CELLS

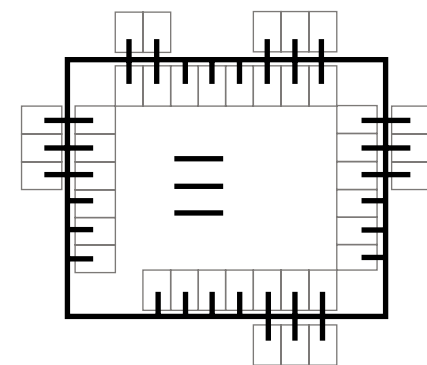
Inmate Housing



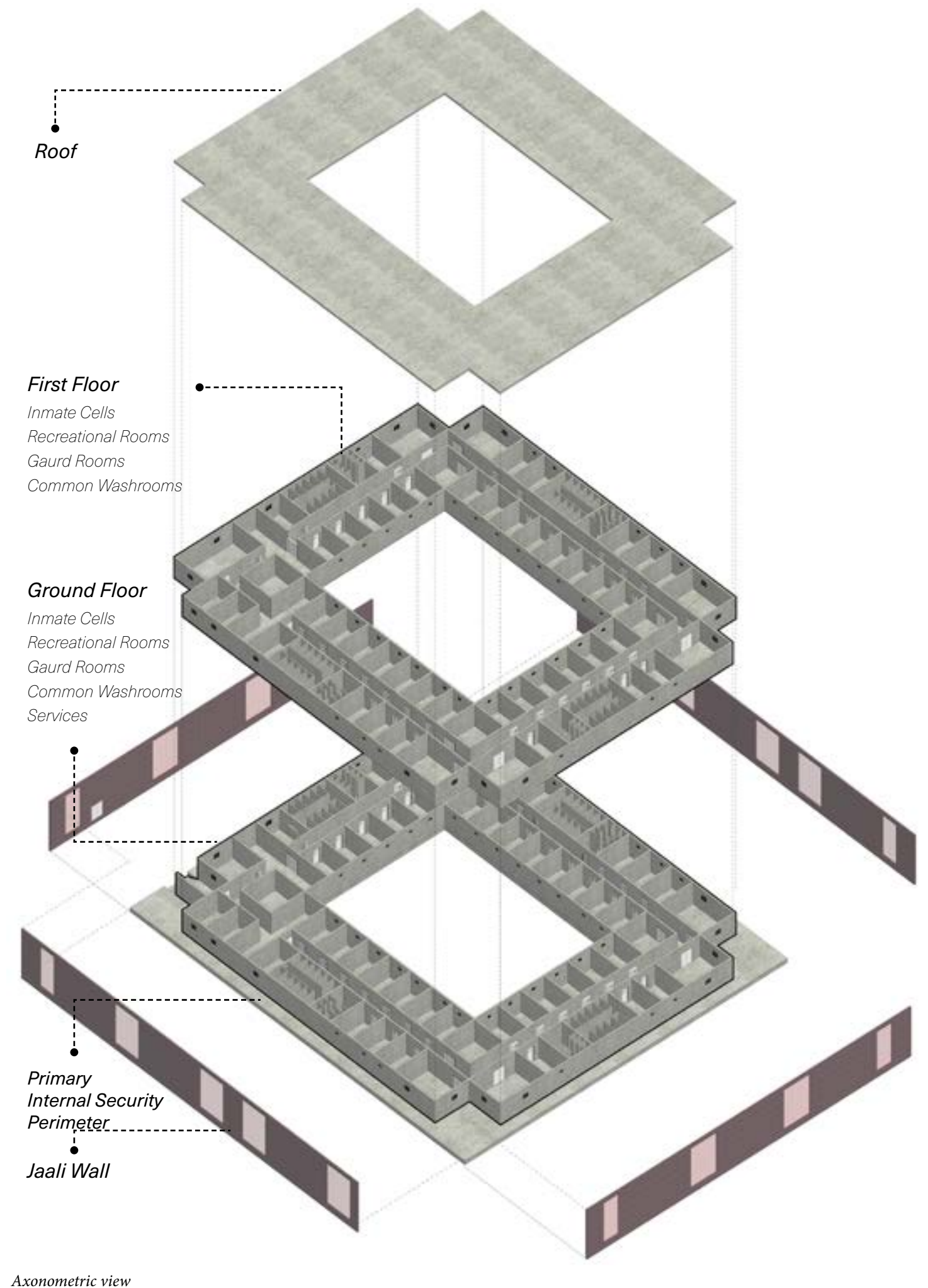
Placement of Watch Towers

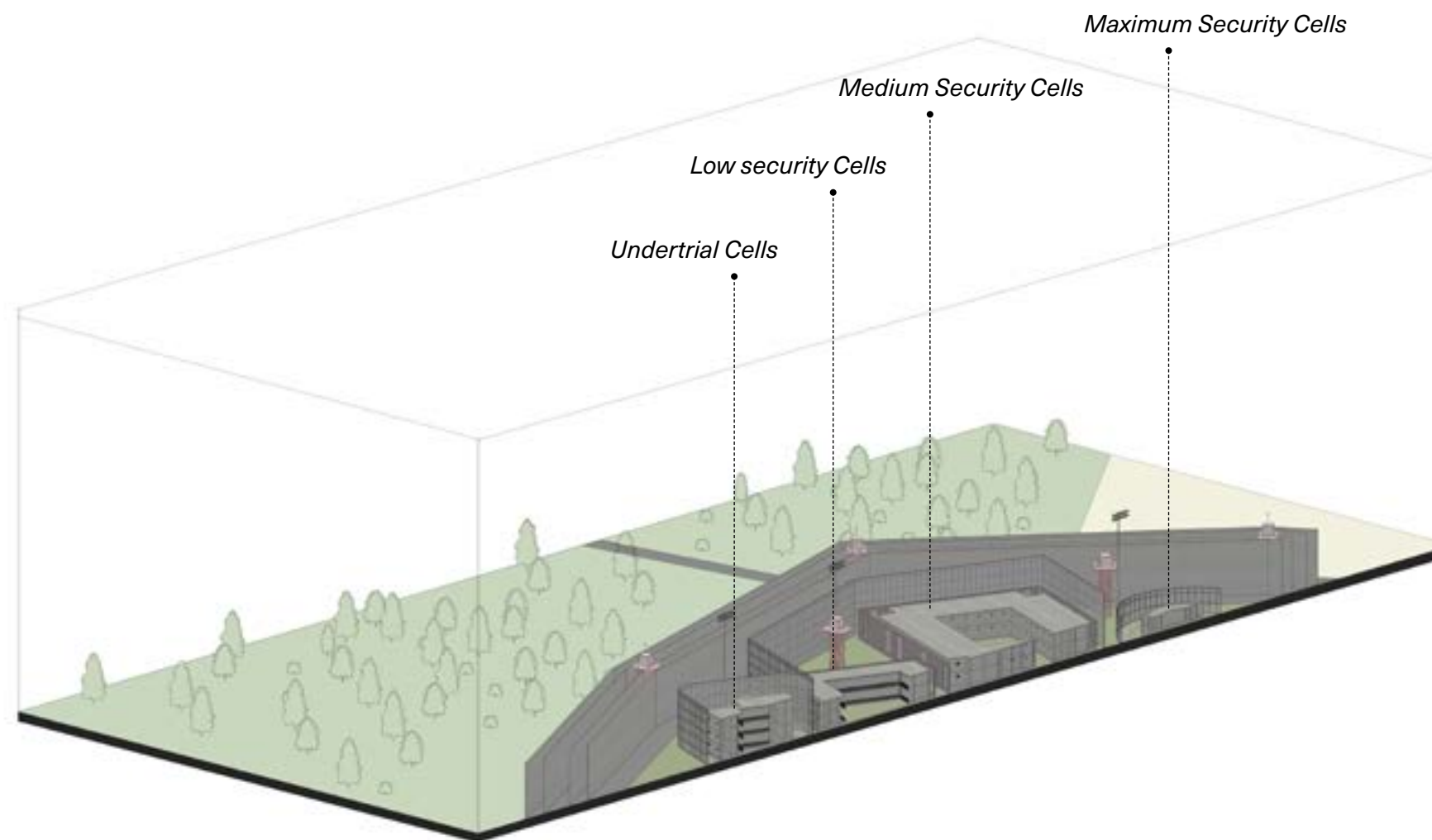


Placement of Residential Units



Residential Unit Layout





Site Section through the inmate housing



View of the prison from outside

06 Thesis Learning Outcomes

The thesis design project explores the arrangement and interaction of spaces within a secure facility which would confine the prisoners as well as liberate them allowing them to repent and process their mistakes.

- The project also enabled me to further develop my skills in architecture and the drive to design for social change.
- The topic chosen was to highlight problems in penal architecture and provide suggestions through changes in design strategies and the architecture program.
- Collectively I hope that the thesis project can contribute to the existing body of knowledge on how architecture can improve the overall quality of a prisoner's life.